GitHub repository where all documentation & code can be found. billbill100 (github.com)

Sensor & Laser Enclosures & Base V1.2 10/03/2026

To print & assemble the sensor enclosure.

The sensor & Laser parts can be printed. Whilst each plate shows multiple parts, to group items together and reduce the number of 3d print files, it is best to print just one part at a time & remove from the build plate before printing the next.

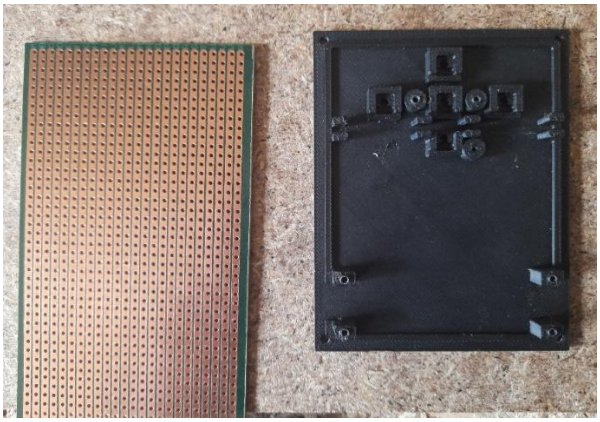
Photo shows sensor & Laser enclosures and base
(not shown, cable clamps & Laser aligning tool)

Note: The 3d printed parts will accept placement of sensors and Lasers for horizontal vertical shutter testing. However this document describes only how to make a horizontal tester.

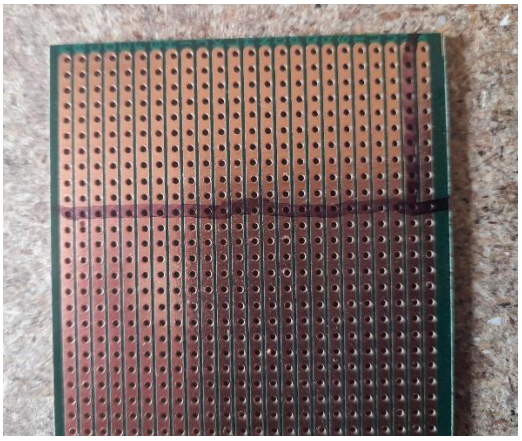
Make the sensor board

The sensors will not fit into the enclosure when mounted on the modules. So a new part is made using stripboard, as detailed below.

The stripboard is cut to fit in the box. Do not cut it to fit too tightly, a little play makes it easier to align. A junior hacksaw will cut the board.



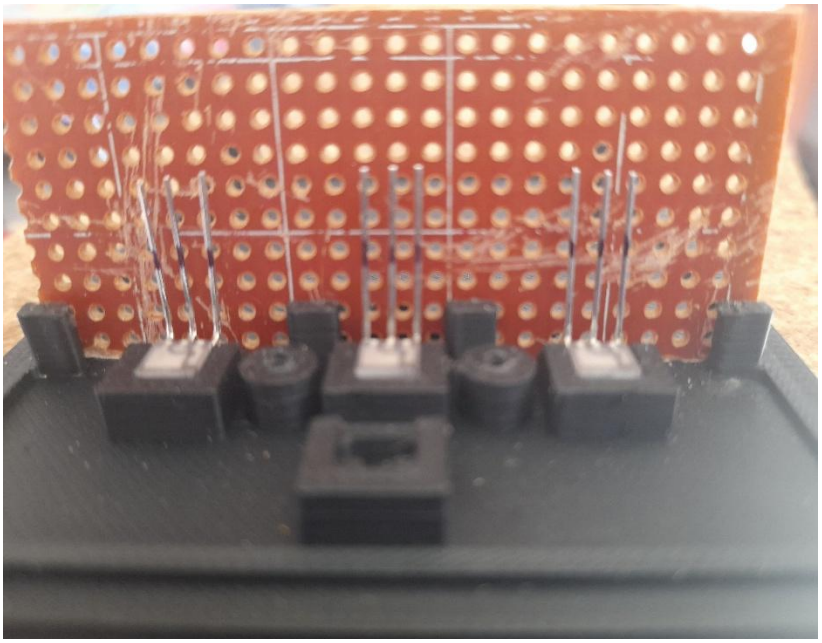
Stripboard and enclosure



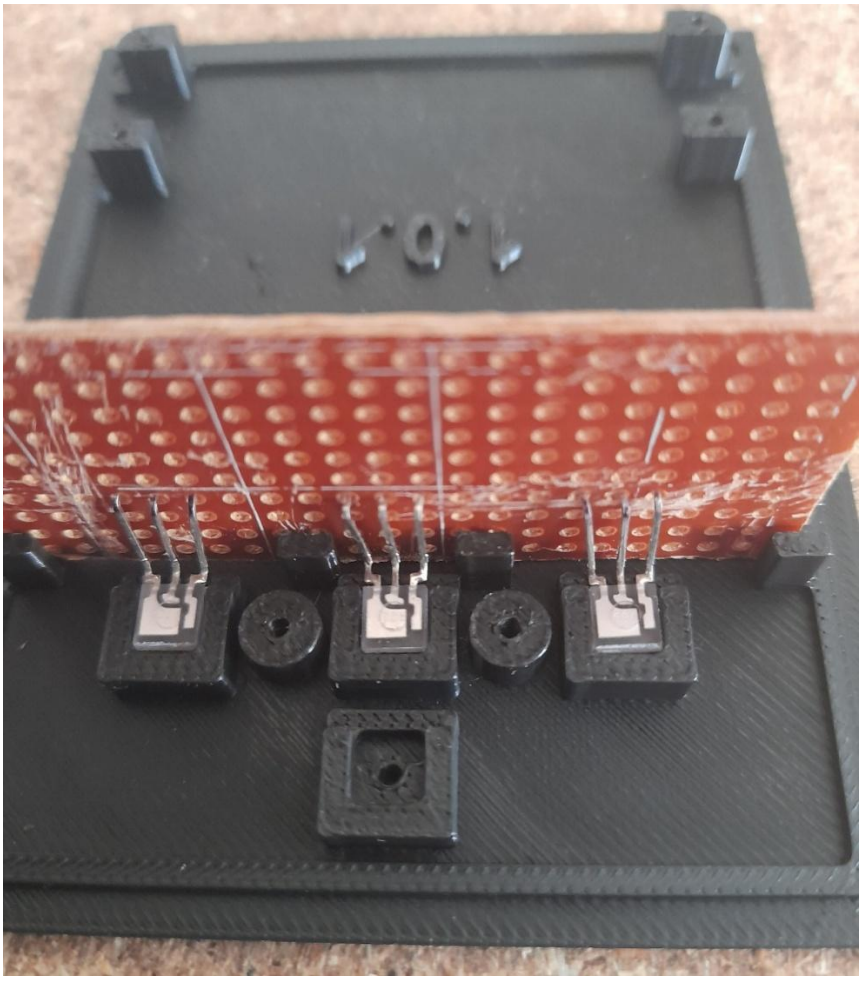
If using the board specified in the parts list, cut through the 11th line of holes then trim of 1 ½ holes from the edge. A junior hacksaw works well for this. The board is tough.

Trial fit the stripboard in the lid and then put into the box, to ensure it is not too long.

CAREFULLY bend the legs of the sensors up and insert them into their holders. (Only two sensors for the Arduino version, in the outside holders).
Mark the legs with a marker-pen where they line up with the holes – black pen used here.



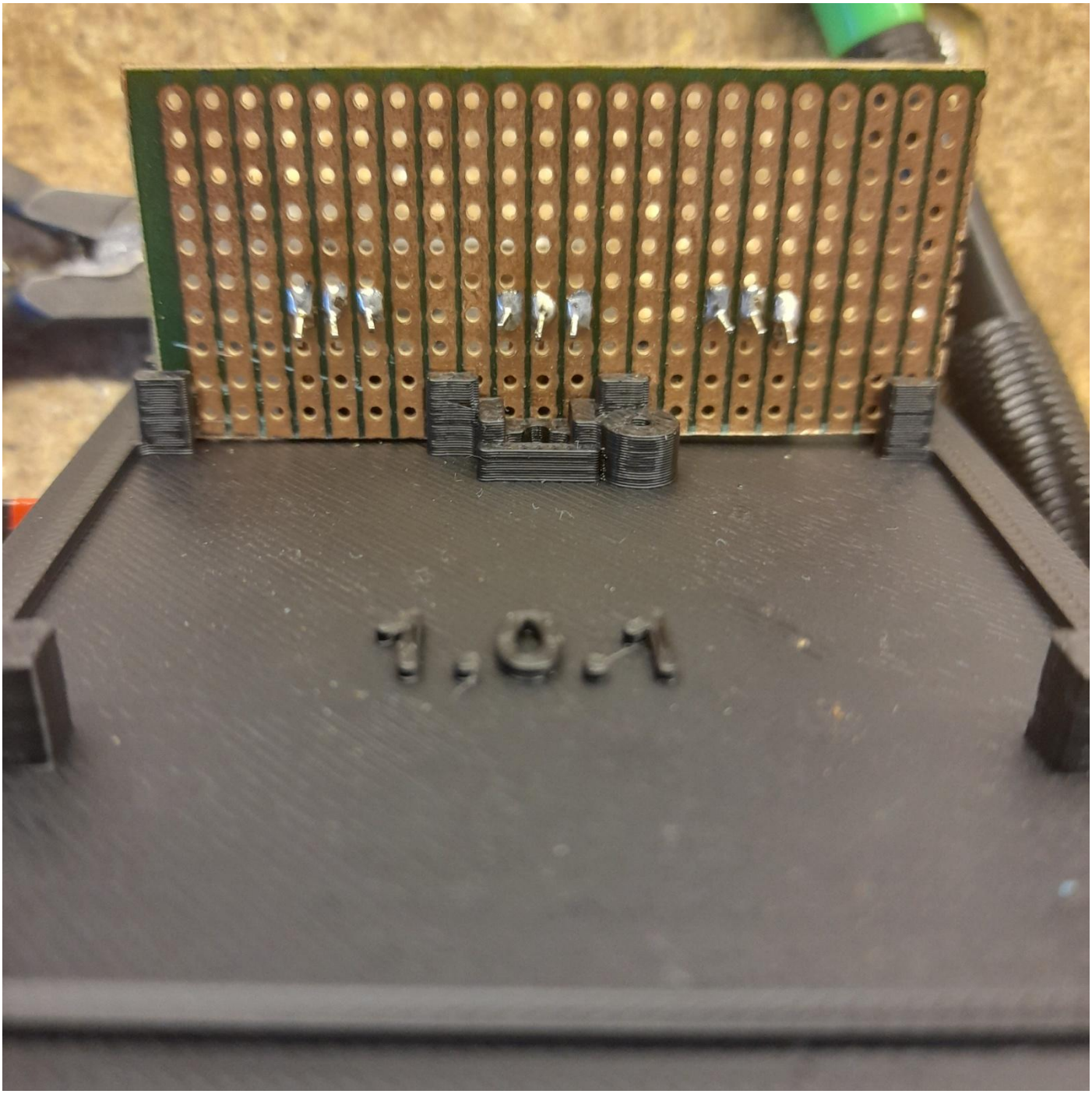
Remove the stripboard and sensors. Bend the legs at the marked point back 90 degrees towards the stripboard. The legs can be placed in the stripboard holes, they will need to be bent apart slightly, then put back into the lid.



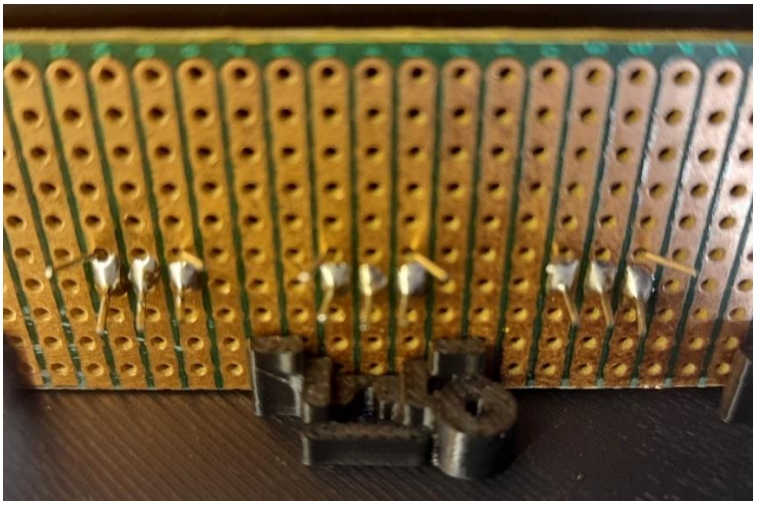
Ensure the sensors are located correctly. By carefylly holding the sensors and turning the lid over, the sensor lens can be seen in the holes.

Fit the T shaped sensor holder and screw into place. (no photo of this step).

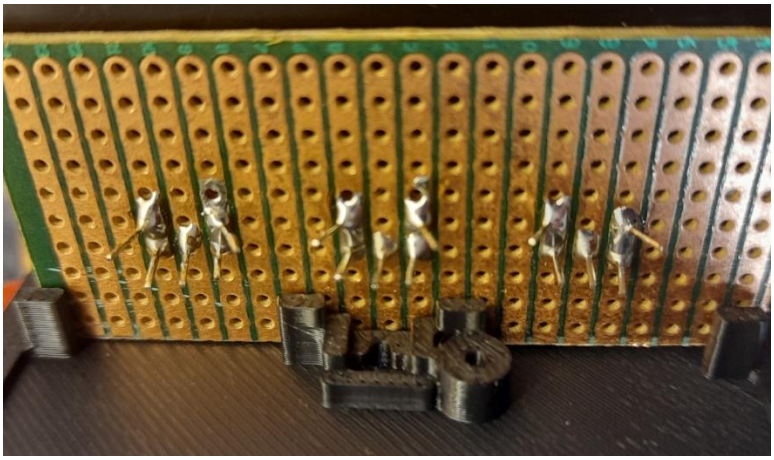
Carefully solder each of the sensor legs. Inspect the solder joints to ensure they have not bridged the copper strips. Using a multimeter on the Ohms scale, measure between each row of copper strips to ensure there are no short circuits.



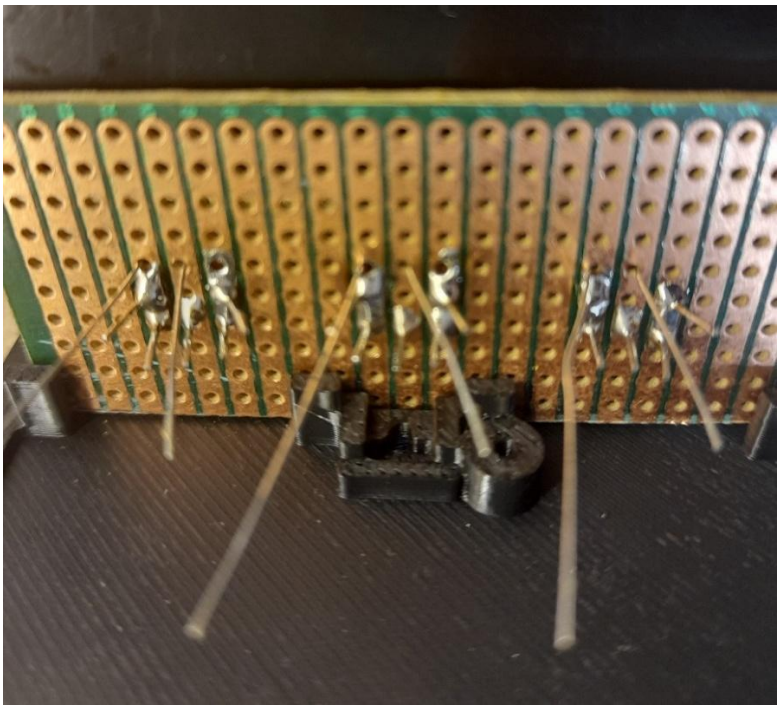
0.1uF capacitors are added. These go across the outer legs of the sensors.



The capacitors can now be soldered.
Visually inspect the soldering and again, using a multimeter, ensure there are no short circuits.

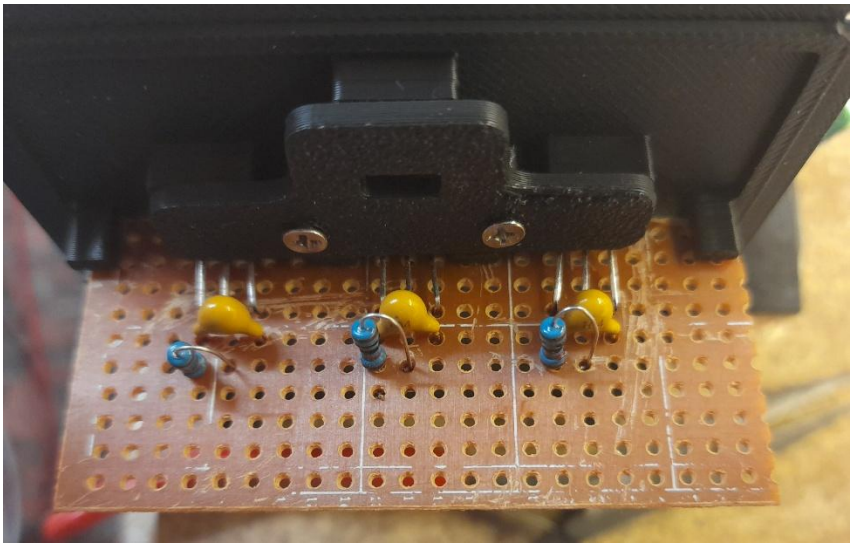


Next 10K resisters are added between the +V and the middle leg of the sensor and soldered.

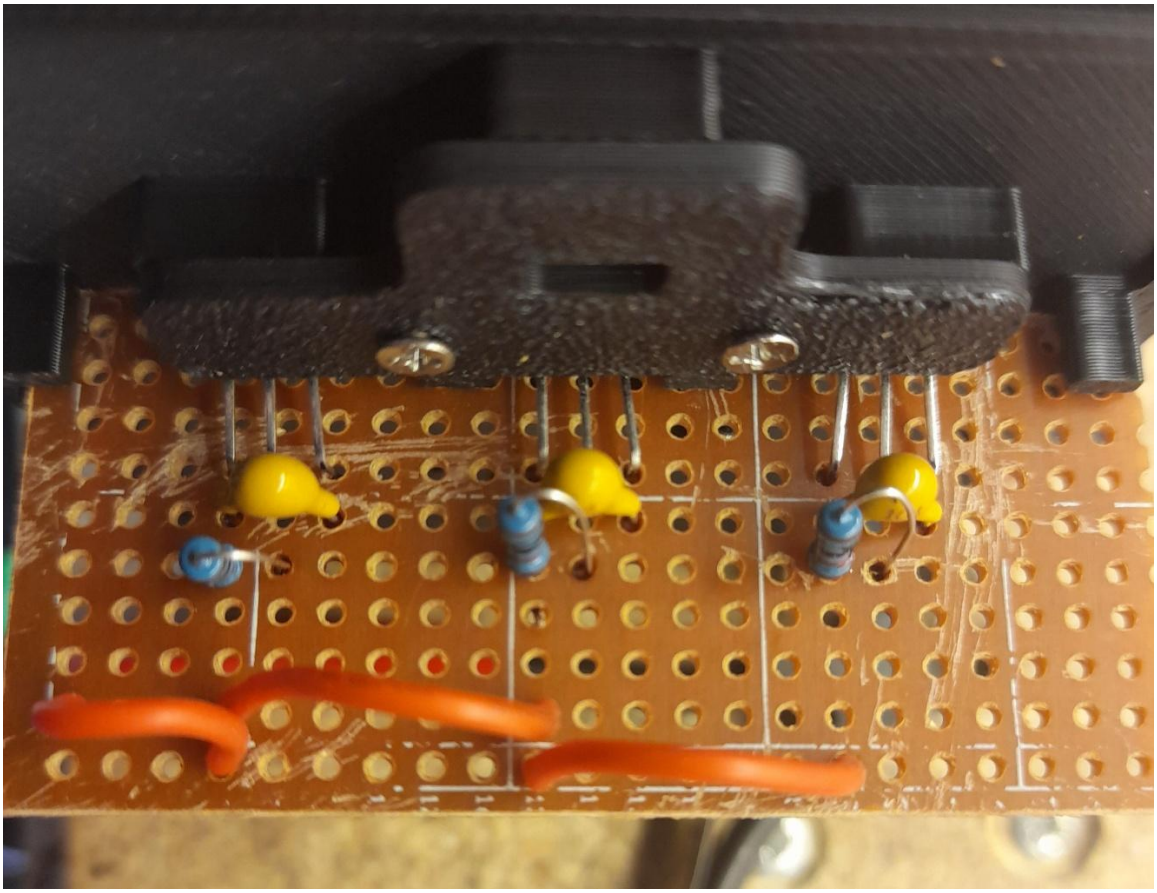


The board will now look like this, with all the components soldered.

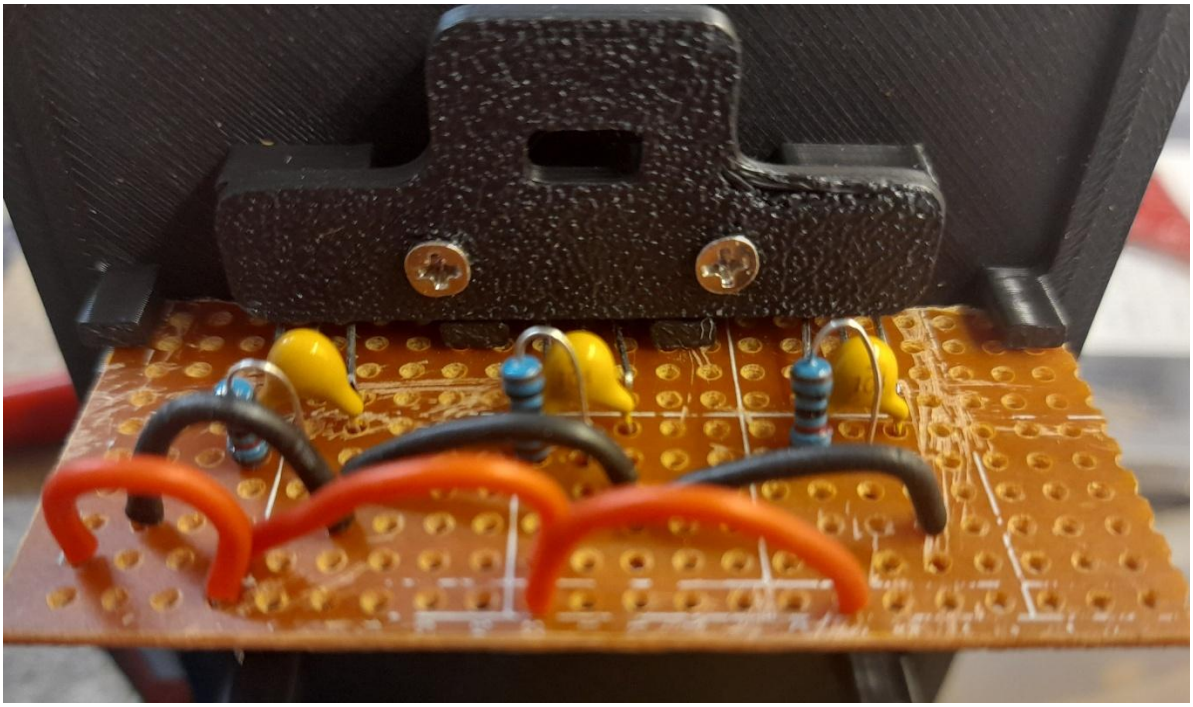
Visually inspect the soldering and again. Using a multimeter, ensure there are no short circuits.
The legs of the components can be trimmed.



The +V of each sensor (left leg) is connected by soldering a wire in daisy-chain, right to left and terminating on the first row of the stripboard.

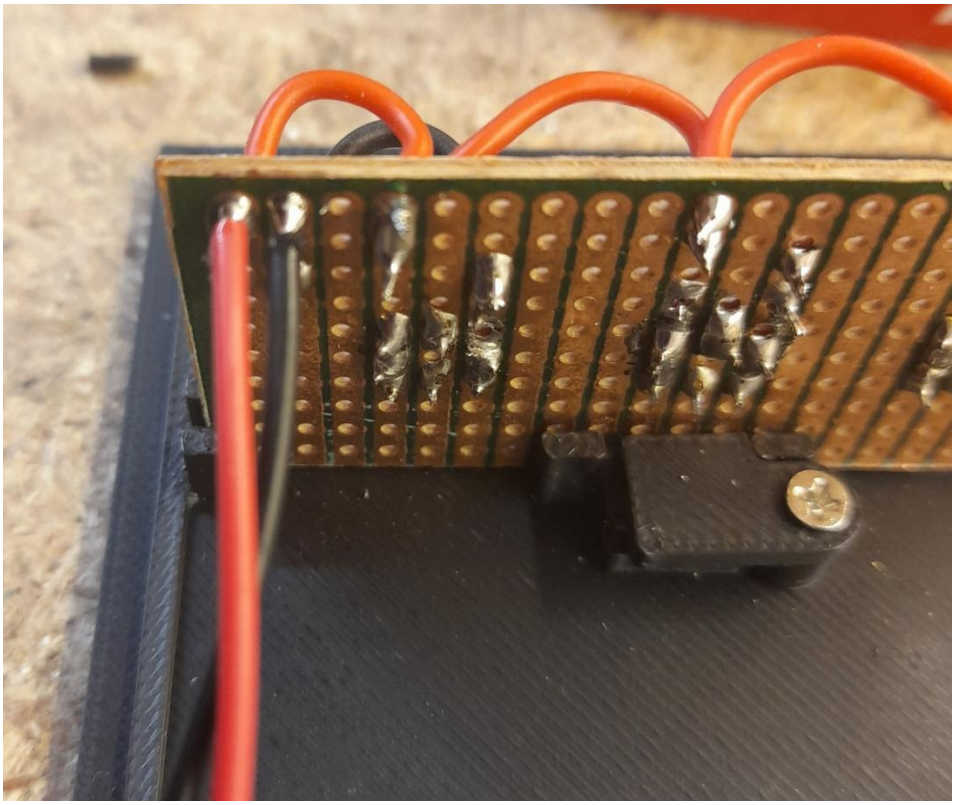


The 0V of each sensor (right leg) is connected by soldering a wire in daisy-chain, right to left and terminating on the second row of the stripboard.

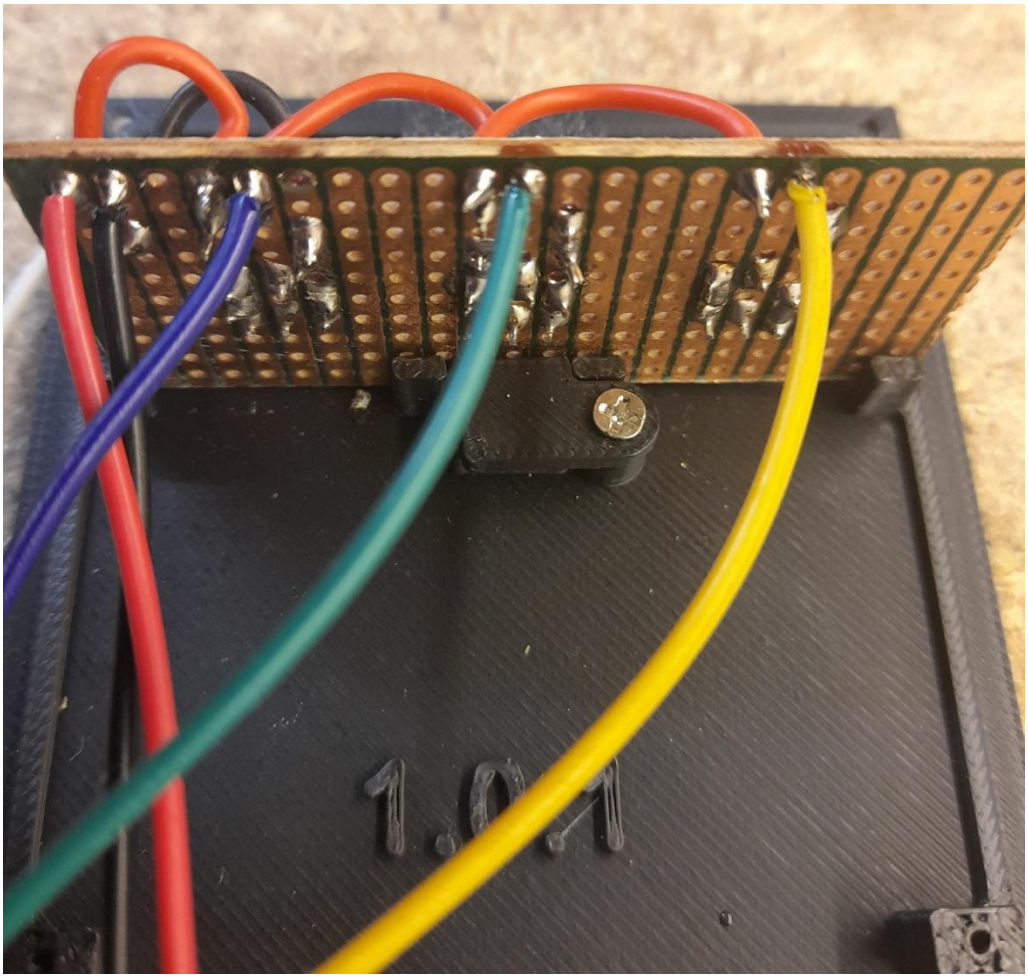


+V & 0V wires from the screw terminal board are now soldered to the respective copper strip.
Note: - on the prototype it was thought a good idea to have wires connected on the bottom, as shown here.
It does work but is more difficult than if they come from above.
It might be better to have the wires entering from above and cutting a corner off the stripboard, to allow the wires through and into the bottom of the box, to exit.

The +V and 0V for the Lasers can also be connected here.



The three wires for the sensors are now soldered to the centre leg of each sensor.



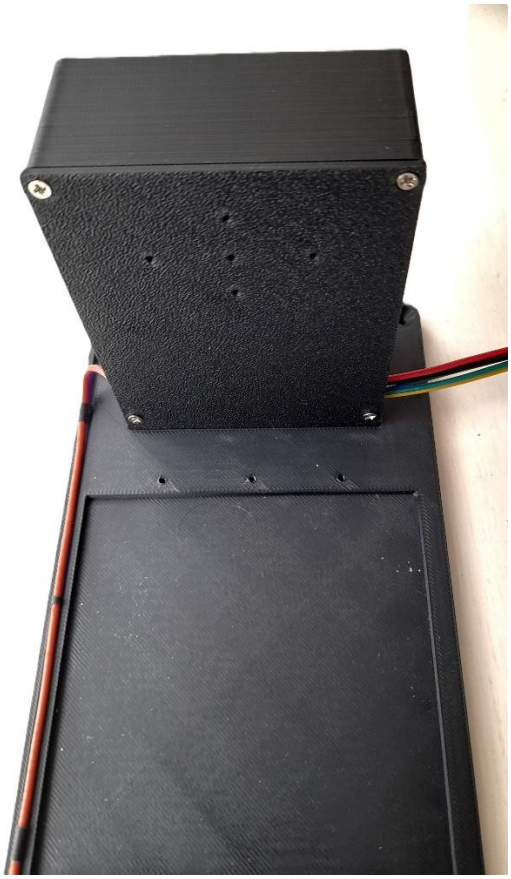
Wires are now secured under the clamps.
The wires can exit either side of the box, depending on user preference.
(White wire shown is not connected. Brown & purple wire on the right feeds the Lasers.)

With a multimeter set on the Ohms scale. Measure between the +V and 0V wires. Ensure there is not a short circuit. Now measure between the +V wire and each of the sensor output wires. Ensure there are no shorts. The meter should show 10K Ohm.
If voltage is applied directly into the microcontroller pin, without the resister, it WILL destroy the board.

To mount the enclosures.
The Laser encosure and base should be printed.

The sensor enclosure is mounted to the base first, using M3 X 12mm bolts, nuts & washers. The base has a slot for running the wires from the sensor enclosure to the Laser enclosure. The slot can be on the left or right, depending on how the builder chooses.
Ensure the enclosure is pushed fully back within the mounting recess, before tightening the bolts, to leave space for the lid.

The lid, containing the sensors can now be fitted, using 2mm x 6mm screws.
Note the wires for the Laser box running in the slot to the left.



The lifting table can now be fitted. The adjusting knob should be on the left, at the bottom.
Two 3d printed clamps fit either side of the table and are fixed with M2 x 8mm screws.

The Laser enclosure is fitted next, using M3 X 12mm nuts, bolts and washers.
Note the orientation of the box. Before tightening the bolts, ensure the box is pushed fully back within the recess, to allow space for the lid.

To assemble & align the Lasers.

The wires soldered onto the Lasers are very thin and delicate. Adding some hot-glue on the solder-joint and onto the insulation will help add strength & prevent them breaking during the next step.

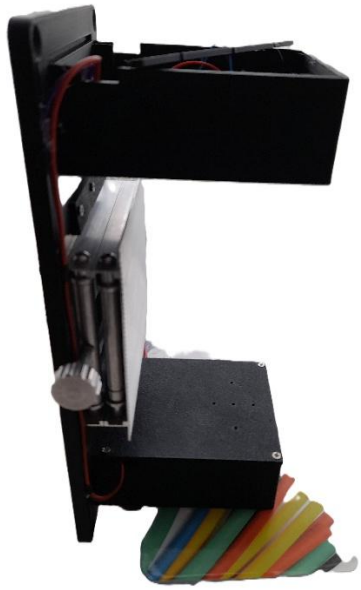
The ends of the Laser +V wires (Red) can be soldered together and connected to the +3 or +5V supply. Do the same with the 0V wires (Blue). Ensure all joints are insulated.

The unit mut now be powered, to illuminate the Lasers.

DO NOT directly look into the Laser or point it your or anybody's eyes.

The Lasers can be focussed, by turning the collar on the end. However, do not focus to a pin-point, the Laser must cover at least the area of the sensor hole. Making the beam too narrow will make alignment very difficult.

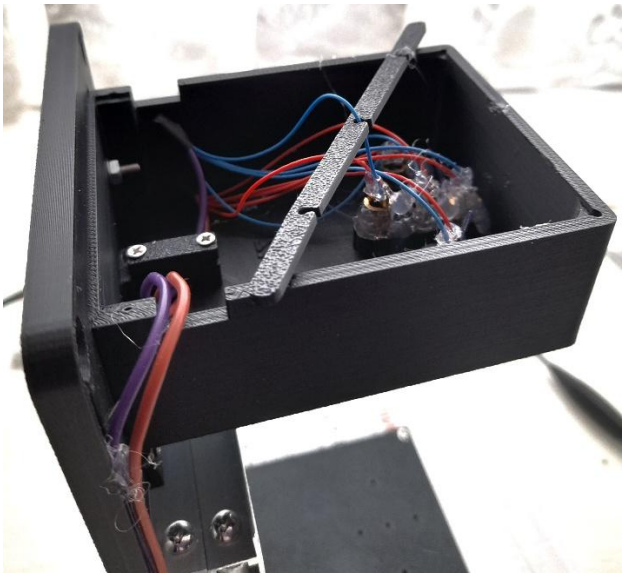
Each Laser needs adjusting to point at the corresponding sensor.
The mounting frame can be turned on-end, so the Laser points down. This makes alignment easier.



Hot-glue is used to hold the Laser in place.
The adjusting tool, pictured below is used to move the wires, which moves the Laser and then hold them in the correct position whilst the glue cools. A small weight can be added to the tool, if it will not stay in position.
Once the glue has cooled, additional hot-glue is added around the Laser.

Do this to each Laser in turn. Patience is required here, to allow the glue to fully cool, before moving onto the next Laser.

Note the photo shows the cables clamped, but do not fit the clamp this until all Lasers are aligned.

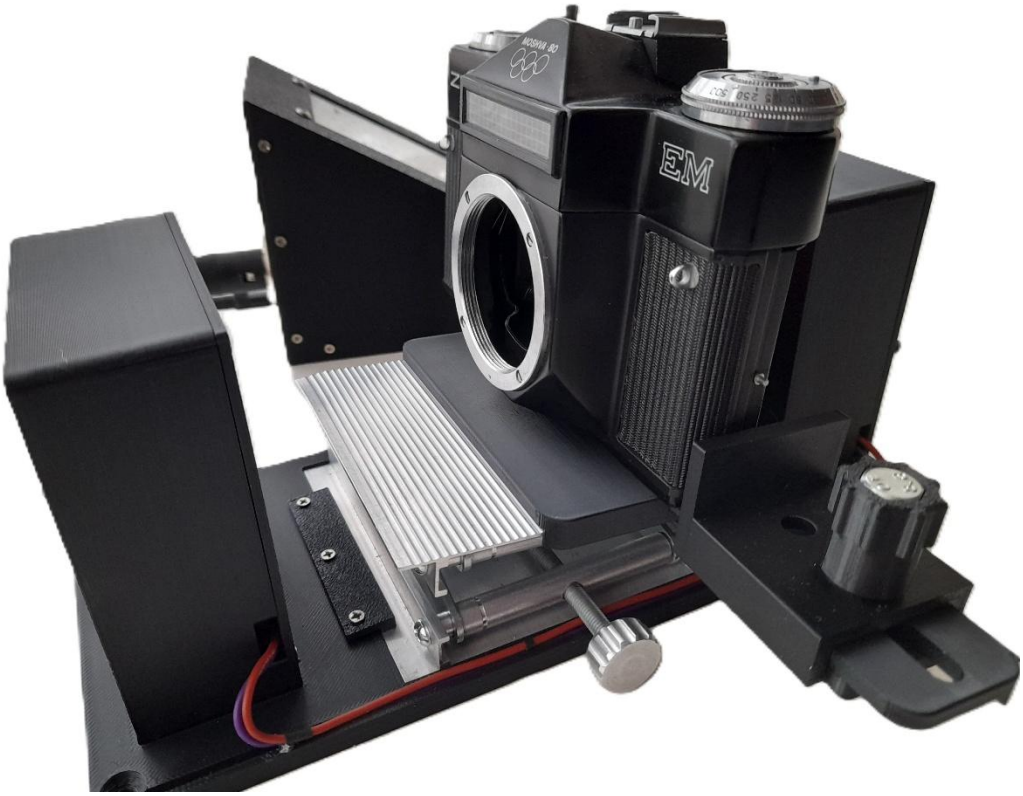


The rear cover can be screwed to the Laser box and this completes assembly.

A further document describes how to print and assemble the camera steady for the adjustable table.

Photos below showing the completed The Shutter Tester.

Sens2 0.000
SensM 0.000
Sens1 2.171
Sens2
SensM
Sens1 1/461
Ready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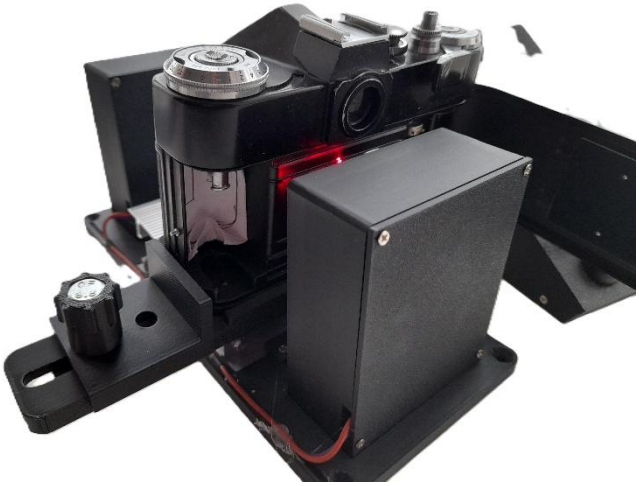

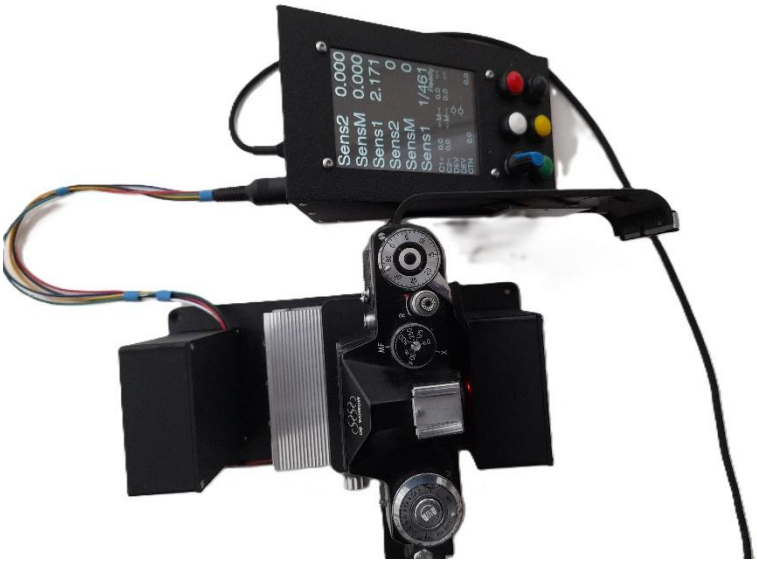
Sens2 0.000
SensM 0.000
Sens1 2.171
Sens2 0
SensM 0
Sens1 1.461
MF